



School of Information Technologies
Faculty of Engineering & IT

SOFT2412/COMP9412 Assignment

Agile Software Development Practices

Unit of Study: SOFT2412/COMP9412

Sprint Number: Sprint 3

Tutorial time: Friday 2pm-4pm

Tutor name: Amyra Meidiana

DECLARATION

We, the undersigned, declare that the work contained in this assignment/project is our own work and has not been copied from other sources or been previously submitted for award or assessment. We understand that failure to comply with the Student Plagiarism: Coursework Policy and Procedure can lead to severe penalties as outlined under Chapter 8 of the University of Sydney By-Law 1999 (as amended). These penalties may be imposed in cases where any significant portion of my submitted work has been copied without proper acknowledgement from other sources, including published works, the internet, existing programs, the work of other students, or work previously submitted for other awards or assessments. We realise that we may be asked to identify those portions of the work contributed by each of us and required to demonstrate our individual knowledge of the relevant material by answering oral questions or by undertaking supplementary work, either written or in the laboratory, in order to arrive at the final assessment mark.

Project Team Members		
Student Name	Student ID	Signature
Zhiwei Cheng	540249451	Zhiwei Cheng
Xuejian Fang	530313995	Xuejian Fang
Boning Zhao	530027564	Boning Zhao

Project Team Members		
Student Name	Student ID	Signature
Xuanhao Yu	530349673	Xuanhao Yu

1. Project Overview

1.1 Project Goals and Objectives

Project goal

Build the Virtual Scroll Access System (VSAS): a Java (Java-17) application that lets users register/login, browse and manage “digital scrolls” (binary files). Guests browse read-only, authenticated users to upload and manage “scrolls”, and administrators to manage users and review usage. The project follows Scrum with time-boxed sprints and CI support, so the client receives a repeatable CI backed release with every increment. Deliver a secure and role aware CLI that lets Edstemus library staff control access to binary scrolls without a graphical interface barrier.

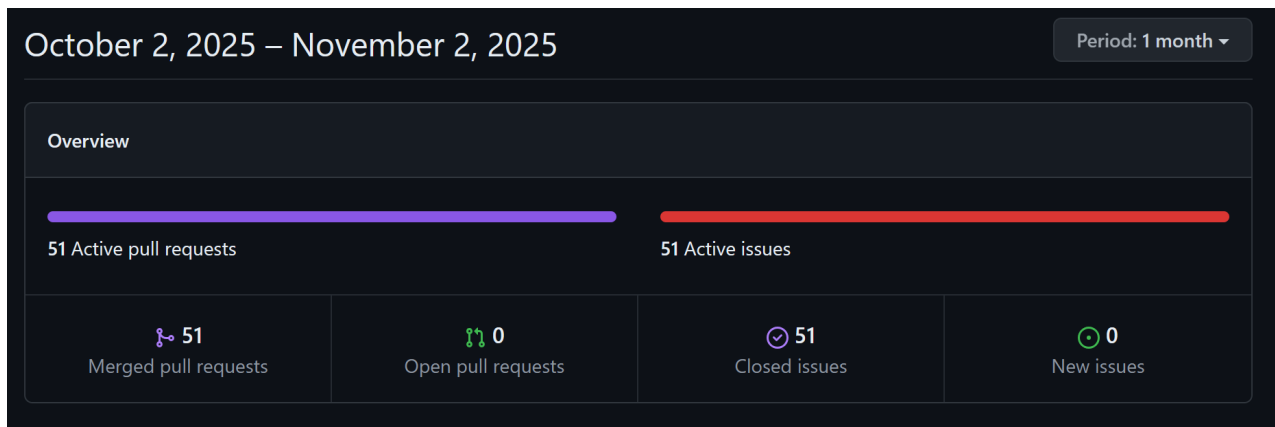
Objectives set for the project

- Implement authentication, session handling, and administrative workflows that satisfy the Virtual Scroll Access System.
- Maintain automated verification across critical commands and repositories to ensure regressions are caught inside the sprint timebox.
- Package and document a runnable artefact that clients can execute with `java -jar` alongside clear operational guidance.

Evidence of features implemented and objectives met

```

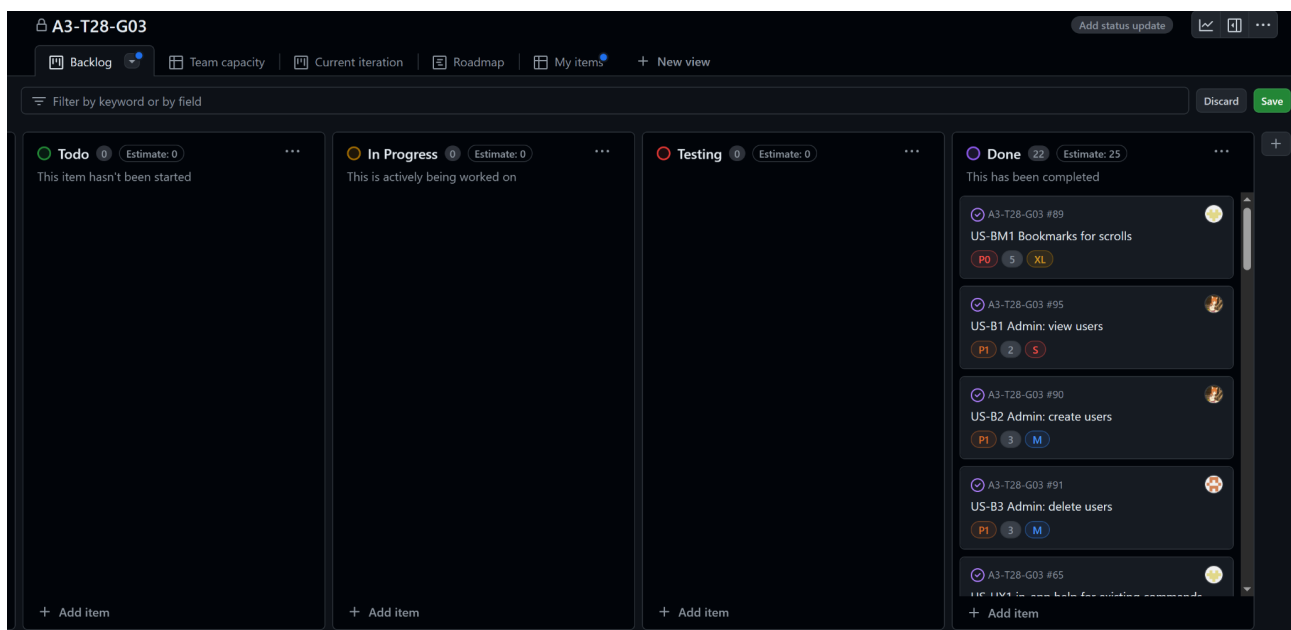
user@sahara:~$ java -jar vsas-3.8.3.jar
Welcome to VSAS CLI. Type 'help' or 'help <command>' for assistance.
vsas> help
Available commands:
  register - Register a new user
  login - Log in with username and password
  logout - Log out of the current session
  whoami - Show the current authenticated user or guest
  list - List scroll metadata with optional filters
  upload - Upload a scroll binary and metadata
  download - Download a scroll to the local filesystem
  bookmark - Manage personal scroll bookmarks
  scroll - Manage scroll lifecycle commands
  preview - Preview metadata and a snippet of a scroll
  profile - Manage profile details
  help - Show help for commands
vsas> help scroll
Command: scroll
Description: Manage scroll lifecycle commands
Usage: scroll <subcommand> [options]
Required flags:
  <subcommand> : One of: delete, update
Optional flags:
  delete : Delete a scroll you uploaded: scroll delete --id <sid> [--yes]
  update : Update metadata or payload: scroll update --id <sid> [--name "<n>"] [--file <path>] [--yes]
Exit codes:
  0 -> success (subcommand dependent)
  2 -> usage error (missing or unknown subcommand)
Example:
  scroll delete --id S-001
vsas> logout
Logout success
vsas> quit
user@sahara:~$
  
```

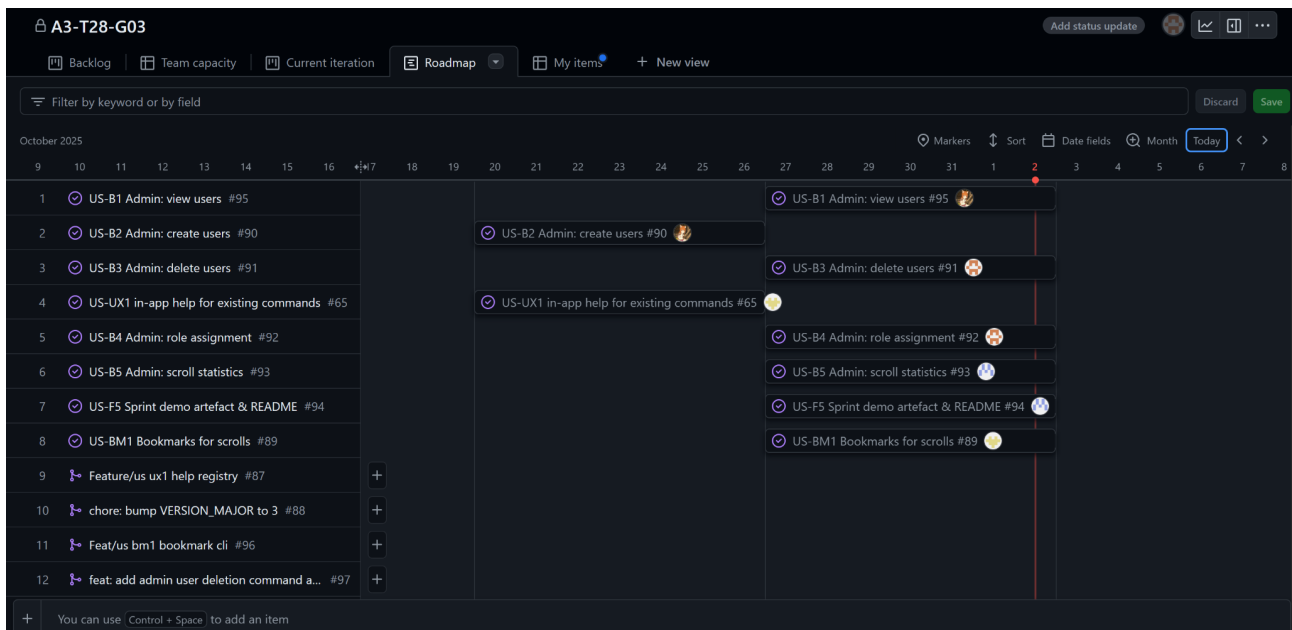


1.2 Sprint Goals

- Finalise the administrative command set so product owners can manage users and statistics through the CLI.
- Introduce personalised bookmarking to improve day to day usability for authenticated readers.
- Produce the Sprint 3 distributable and supporting documentation through the CI toolchain.
- Evidence:
- Sprint 3 GitHub Project board cards for US-B1, US-B2, US-B3, US-B4, US-B5, US-BM1, US-F5 and US-UX1 moved to Done with merged pull requests ([#98](#), [#100](#), [#102](#), [#97](#), [#99](#), [#96](#), [#105](#), [#87](#)).
- BookmarkAddCommand.java, BookmarkListCommand.java, and BookmarkRemoveCommand.java shipped alongside BookmarkCommandsTest.java and FileBookmarkRepositoryTest.java.
- Jenkins build logs and artefacts confirm publication of the fat JAR renamed to vsas-cli.jar, consistent with the guidance in README.md.

Evidence from the project board (Sprint-3).





1.3 User Stories

User stories in Sprint-3

US code	US-B1	US-B2	US-B3	US-B4	US-B5	US-BM1	US-F5	US-UX1
issue #	#95	#90	#91	#92	#93	#89	#94	#65

User stories considered in Sprint-3.

- **US-B1 Admin:** view users to provide filtered visibility of user accounts.
- **US-B2 Admin:** create users to streamline onboarding through CLI prompts.
- **US-B3 Admin:** delete users with safeguards against self removal and scroll ownership conflicts.
- **US-B4 Admin:** role assignment to manage privilege transitions.
- **US-B5 Admin:** scroll statistics for oversight of per scroll and per uploader activity.
- **US-BM1:** Bookmarks for scrolls to let authenticated users curate personal lists.
- **US-F5** Sprint demo artefact and README to ship the runnable CLI bundle with instructions.
- **US-UX1** in-app help for existing commands

Changes during the sprint.

Hardened TSV path resolution for admin commands (commits [d614121](#), [07e964a](#)) to accommodate Gradle execution contexts.

Updated scroll counters after downloads (commit [de0a4f7](#)) so analytics remain consistent with repository state.

Corrected bookmark repository header handling (commit [507b24a](#)) to keep duplicate detection reliable.

2. Scrum Artefacts and Methodologies

2.1 Sprint Planning

Sprint Date: Week 12

Sprint Duration: 1 week

Sprint Goal:

The third sprint, conducted during Week 12, aimed to extend the functionality of the **Virtual Scroll Access System (VSAS)** by introducing additional administrative and user convenience features, improving usability, accessibility, and system monitoring based on feedback from the previous sprint. The sprint planning meeting was held at the start of the week, where the team discussed development priorities, assigned user stories, and refined the existing backlog to ensure continuous and incremental delivery.

Boning Zhao, serving as the *Product Owner*, was responsible for generating new user stories, assigning tasks to team members, creating issues on GitHub, and managing the overall storyboard. In addition to coordination work, Boning implemented the **Bookmarks for Scrolls** feature, allowing users to save and quickly access frequently used scrolls, and developed the **In-App Help** module to display usage instructions for existing CLI commands. These features collectively improved user convenience and system usability.

Xuejian Fang, acting as the *Scrum Master*, implemented and tested the **Admin View Users** and **Admin Create Users** functionalities. He also set up automated build and release processes within the Jenkins pipeline, enabling the system to automatically build the JAR file, generate version numbers, and publish corresponding GitHub releases and tags. In addition, Xuejian configured code review rules that require each pull request to be approved by at least one reviewer before merging. He maintained and deployed the team's Jenkins server and added testing visualisation plugins such as the Coverage plugin for report generation and review.

Xuanhao Yu was responsible for implementing the **Admin Scroll Statistics** functionality, enabling administrators to view upload and download metrics for each scroll. He also prepared the **Sprint Demo Artefact** and updated the **README** file with detailed setup instructions and sprint summaries. In addition, Xuanhao conducted testing to verify data consistency and assisted in documentation preparation for sprint evaluation.

Zhiwei Cheng implemented the **Admin Delete Users** and **Admin Role Assignment** functionalities, allowing administrators to remove user accounts and modify their roles securely. He conducted unit and integration testing to ensure data integrity and proper enforcement of access control rules. Zhiwei also participated in code review and validation of administrative operations to ensure repository consistency and maintain project quality standards.

2.2 Daily Stand-ups

All stand-up meetings for this sprint were held in person, continuing the efficient communication pattern established in previous iterations. The first meeting took place on Monday, marking the start of **Sprint 3**. During this session, the Product Owner assigned the new user stories to each team member and reviewed the development plan for the week. With the experience gained from earlier sprints, the team followed the same structured approach for task allocation, issue tracking, and progress documentation to ensure consistent workflow and delivery quality. This meeting lasts for 10 minutes.

The second meeting was held later in the week, where all members jointly tested the newly implemented functionalities and merged their completed features into the main branch. This collaborative testing session focused on verifying that all administrative and user-oriented commands—such as role assignment, deletion, and statistics—worked correctly after integration. Any minor issues discovered were fixed on the spot to ensure a stable final build. This meeting lasts for 10 minutes..

Overall, the sprint meetings maintained effective coordination and transparency, enabling the team to consolidate their progress, validate the new features, and complete Sprint 3 according to the planned objectives.

The meeting photo is in the link below:

https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/tree/Appendix_A

2.3 Retrospective

The sprint retrospective meeting was held at the end of Week 12, with all team members present. The discussion focused on reviewing the overall progress of Sprint 3, evaluating the completion of assigned user stories, and identifying both technical and collaborative improvements for the next iteration.

The team successfully achieved all major objectives for this sprint. Core functionalities, including Admin Role Assignment, Delete Users, View and Create Users, Scroll Statistics, Bookmarks, and In-App Help, were fully implemented and verified through unit and integration testing. These new features significantly enhanced the system's administrative management and user accessibility. The continuous integration and automated release processes set up in previous sprints continued to function smoothly, ensuring that the updated version was automatically built, tagged, and released via Jenkins and GitHub.

The quality of collaboration and workflow consistency also improved, as team members were now familiar with the established development process. Code reviews were conducted regularly to ensure coding standards and prevent potential conflicts, and testing coverage remained stable through automated reporting and Jenkins visualisation plugins.

Despite these successes, the team identified several opportunities for further improvement. The command-line interface could be made more interactive and user-friendly, especially for first-time users. Some administrative commands could benefit from enhanced validation and clearer error handling. Additionally, minor delays were occasionally encountered in the automated build stages, suggesting the need for additional pipeline optimisation.

2.4 Use of Scrum Artefacts

All Scrum artefacts for Sprint 3, including the Product Backlog, Sprint Backlog, and User Stories, are maintained on the GitHub Projects board. The board records task priorities, assignments, and progress status throughout the sprint.

Project backlog and Sprint backlog:

<https://github.sydney.edu.au/orgs/SOFT2412-COMP9412-2025s2/projects/309/views/1>

The team also improved the visibility of the CI/CD process through Jenkins. Automated builds, testing, and versioned releases were integrated into the Jenkins pipeline, which generated visual test and coverage reports through installed plugins. Evidence of these artefacts and automation outputs can be accessed via the team's GitHub repository and Jenkins interface:

The Jenkins Console Output:

https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/blob/Appendix_A/jenkins_console_wk12.png

3. Agile Development Tools and Practices

3.1 Version Control

3.1.1 Tools and Branching Strategy

Unchanged from last week: Git + GitHub (Issues, Projects, PRs), Gradle builds/tests with google-java-format, and the feature-branch model (feature/<story-name>), conventional commit prefixes, and PR-only merges to a protected main. Branch protection rules on main remain in effect: require a pull request before merging; require status checks to pass (Jenkins pipeline must be green for the PR's head SHA); require at least 1 reviewer approval; block force pushes. The only change this week is in the CI release artifact naming: Jenkins now publishes the auto-built jar as vsas-3.8.3.jar (i.e., vsas-<version>.jar) instead of the previous app-all.jar, embedding the current release version in the filename for clearer traceability.

3.1.2 Merge Conflicts and Branch Integration

Process this week: open a PR from feature/*; Jenkins automatically runs the pipeline (build/tests/format) on the PR's head SHA; merging is blocked until the pipeline is green and at least one reviewer approves; we then merge (typically squash) to keep history clean. No new conflict incidents were recorded.

3.1.3 How Added Features Fit User Requirements

Following user feedback requesting better, more usable in-CLI guidance, this sprint delivered US-UX1 In-app help for existing commands (#65). We implemented a central CommandRegistry to own command metadata and refactored the dispatcher to read from it; added a HelpCommand that lists all commands or detailed per-command usage/flags/examples; printed a one-line REPL banner prompting users to type help or help <command>; and standardised exit codes (0 success, 1 validation/permission, 2 usage, 3 IO) across commands and tests. This meets the discoverability and UX goals without altering existing workflows, aligning the implementation with the story's acceptance criteria and keeping guidance consistent with the actual command definitions.

- Merge pull request #105 from SOFT2412-COMP9412-2025s2/chore/release-rename-jar-to-vavs Cheng Zhiwei Cheng - fix: rename vavs to vsas xfan0282 - chore(release): upload fat jar as vavs--version-jar xfan0282 - Merge pull request #106 from SOFT2412-COMP9412-2025s2/docs/improve-readme Cheng Zhiwei Cheng - docs: add repository URL xfan0282 - docs: improve README with clearer setup and usage instructions xfan0282 - Merge pull request #103 from SOFT2412-COMP9412-2025s2/feat/register-command-role-support Boning Zhao - feat: enhance registration command to support role specification and add tests for admin user login Zche0826 - J CommandRegistry.java app/src/main/java/org/soft2412/vsas/ccli - J RegisterCommand.java app/src/main/java/org/soft2412/vsas/ccli - J LoginCommandTest.java app/src/test/java/org/soft2412/vsas/ccli - J RegisterCommandTest.java app/src/test/java/org/soft2412/vsas/ccli - Merge pull request #102 from SOFT2412-COMP9412-2025s2/feature/us-b2-admin-add-users Xuejian Fang - J AdminUsersAddCommand.java app/src/main/java/org/soft2412/vsas/ccli - J CommandRegistry.java app/src/main/java/org/soft2412/vsas/ccli - J LoginCommand.java app/src/main/java/org/soft2412/vsas/ccli - J FileUserRepository.java app/src/main/java/org/soft2412/vsas/repo - J AdminUsersAddCommandTest.java app/src/test/java/org/soft2412/vsas/ccli - fix(repo): correct default users TSV path resolution xfan0282 - style: format all Java sources with Google Java Format xfan0282 - fix(login): read users via FileUserRepository to honor unified path resolution xfan0282 - test(ccli): cover admin users add (success, duplicates, invalids, permission) xfan0282 - feat(ccli): register 'admin users add' in CommandRegistry with help/usage xfan0282 - feat(ccli): add AdminUsersAddCommand for creating users (admin-only) xfan0282 - Merge pull request #99 from SOFT2412-COMP9412-2025s2/feat/admin-stats Xuanhao Yu - fix: resolve merge conflicts in AdminUsersList and AdminStats Xuanhao - test: improve AdminStatsCommand coverage to 93% (branches 76%) via invalid arg & shorten() reflection tests Xuanhao - Add AdminStatsCommand integration tests; coverage 67% — verify per-scroll and uploader summary tables, permission and empty cases Xuanhao - fix: update download/upload counters on scroll actions Xuanhao - US-B5: add admin stats command ,now can view uploader, upload and download counts Xuanhao - Merge pull request #100 from SOFT2412-COMP9412-2025s2/feature/us-b1-admin-users-list Xuejian Fang - style: format all Java sources with Google Java Format xfan0282 - fix(ccli): prefer project-root data/users.tsv over app/data/users.tsv xfan0282 - fix(ccli): make 'admin users list' find data/users.tsv from project root when run via Gradle xfan0282 - test(ccli): AdminUsersListCommand lists all users and supports AND-filters xfan0282 - feat(ccli): register 'admin users list' and add help metadata xfan0282 - feat(ccli): add 'admin users list' command with filters and fixed-width output xfan0282 - Merge pull request #98 from SOFT2412-COMP9412-2025s2/feat/US-B4admin-users-role-command Boning Zhao - feat: implement admin user role update command and repository support Zche0826 - Merge pull request #97 from SOFT2412-COMP9412-2025s2/feat/US-B3admin-users-delete-command Cheng Zhiwei Cheng - feat: add admin user deletion command and repository support Zche0826 - Merge pull request #96 from SOFT2412-COMP9412-2025s2/feat/us-bml-bookmark-ccli Cheng Zhiwei Cheng - test: add unit tests for FileBookmarkRepository functionality bzh8676 - fix: skip header line when reading bookmark data in listByUser and exists methods bzh8676 - test: add unit tests for bookmark commands and repository functionality bzh8676 - feat: implement bookmark management commands (add, list, remove) and associated repository functionality bzh8676 - Merge pull request #87 from SOFT2412-COMP9412-2025s2/feature/us-ux1-help-registry Xuejian Fang - feat: implement access control for commands and update related functionality bzh8676 - test: add tests for command registry and help command functionality bzh8676 - feat: implement command registration and help system for CLI bzh8676 - fix: update error return codes for consistency across commands bzh8676 - Merge pull request #88 from SOFT2412-COMP9412-2025s2/chore/bump-major-version Xuejian Fang - chore: bump VERSION_MAJOR to 3 xfan0282 - Merge pull request #86 from SOFT2412-COMP9412-2025s2/chore/remove-legacy-list-output Boning Zhao - test(list): update tests to verify fixed-width output format bzh8676 - refactor(list): remove legacy output for backwards compatibility bzh8676 - Merge pull request #85 from SOFT2412-COMP9412-2025s2/fix/login-single-prompt-and-usage Boning Zhao - test(login): add tests for single password prompt and fast-fail usage when username missing xfan0282 - fix(login): prompt for password once; require username before prompting and show usage xfan0282 - Merge pull request #84 from SOFT2412-COMP9412-2025s2/feat/us-a3-update-profile Cheng Zhiwei Cheng - Merge branch 'feat/us-a3-update-profile' of https://github.com/sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03 into feat/us-a3-update-profile bzh8676 - test(repo): add test for handling rows with missing trailing columns in profile update bzh8676 - fix(profile): handle missing fields during user profile update in TSV bzh8676 - test(repo): add tests for updating user profile fields and password handling bzh8676 - feat(profile): add profile update command and repository method bzh8676 - test(repo): add test for handling rows with missing trailing columns in profile update bzh8676 - fix(profile): handle missing fields during user profile update in TSV bzh8676 - test(repo): add tests for updating user profile fields and password handling bzh8676 - feat(profile): add profile update command and repository method bzh8676 - Merge pull request #82 from SOFT2412-COMP9412-2025s2/feat/us-e1-track-scroll-usage Boning Zhao - fix: correct comment typo in UTF-8 handling logic Zche0826 - feat: track scroll upload/download counts - US-E1 Zche0826 - Merge pull request #81 from SOFT2412-COMP9412-2025s2/feat/us-d4-preview-scroll Cheng Zhiwei Cheng - fix(dispatcher): add preview case Zche0826 - feat(preview): enhance UTF-8 handling and add tests for text and binary previews Zche0826 - refactor(tests): clean up formatting and remove unused imports in PreviewCommandTest Zche0826 - feat(ccli): add preview command to display scroll metadata and content snippet Zche0826 - Merge pull request #80 from SOFT2412-COMP9412-2025s2/feat/d3-search-filters Boning Zhao - fix(list): print fixed-width table before legacy rows so tests match first 's12' line as fixed-width xfan0282 - fix(list): also print legacy pipe-separated rows to keep backward-compatible tests green xfan0282 - test(list): cover filters, invalid date errors, empty-state message, and fixed-width table stability xfan0282 - feat(list): stable fixed-width table output xfan0282 - fix(list): make captured filter variables final for stream; remove unused import xfan0282 - feat(list): parse filters and validate dates (yyyy-MM-dd) xfan0282 - Merge pull request #79 from SOFT2412-COMP9412-2025s2/feat/g1-secure-password-prompt Boning Zhao - test(ccli): add tests for PasswordPrompt and interactive password flows xfan0282 - fix(register): check non-password required flags before prompting xfan0282 - feat(login): support masked password prompt when --password is omitted xfan0282 - feat(register): prompt for masked password when --password is omitted xfan0282 - feat(ccli): add PasswordPrompt with masked input, fallback, and test seam xfan0282 - Merge pull request #76 from SOFT2412-COMP9412-2025s2/fix/release-feat-detection-anchor Xuejian Fang - fix: update comment Xuejian Fang - fix(release): anchor feat detection to PR source branch and support squash titles xfan0282 - Merge pull request #78 from SOFT2412-COMP9412-2025s2/feat/scroll-update Xuejian Fang - style(ccli): apply Google Java Format for US-C2 scroll update Xuanhao - test(scroll): add ScrollUpdateSubcommand tests for US-C2 update feature (85% instruction and 76% branch coverage) Xuanhao - feat(scroll): implement scroll update command for US-C2 Xuanhao - Merge pull request #75 from SOFT2412-COMP9412-2025s2/fix/release-feat-detection Xuejian Fang - fix(release): simplify feat regex to avoid Groovy escape error xfan0282 - fix(release): escape parentheses in feat regex xfan0282 - fix(release): detect feat commits from squash merges xfan0282 - Merge pull request #73 from SOFT2412-COMP9412-2025s2/feat/scroll-delete Xuanhao Yu - test(scroll): add ScrollCommand test to cover CLI usage and subcommands Xuanhao - test(scroll): add ScrollDeleteSubcommand test (84% instruction, 70% branch coverage) Xuanhao - style(ccli): update interactive prompt text and reformat code to Google Java Style Xuanhao - feat(scroll): support 'scroll delete --id <id>' for owner-only deletion Xuanhao - Feat/US d2 download scroll (#72) Cheng Zhiwei Cheng - Merge pull request #62 from SOFT2412-COMP9412-2025s2/ci/release-jar Xuejian Fang - feat(ccli): build fat jar with shadowJar and upload it to GitHub Release xfan0282 - feat(build): add Shadow plugin to produce fat/uber JAR with minimal changes xfan0282	main feature/us-b1-a feat/d3-search feat/g1-secure ci/release-jar
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	Merge pull request #61 from SOFT2412-COMP9412-2025s2/fix/coverage-adapter-compat Xuejian Fang	
	fix(cli): drop jacocoAdapter usage, use legacy parser map for recordCoverage and guard it xfan0282	
	Merge pull request #60 from SOFT2412-COMP9412-2025s2/fix/release-gh-headsha-idempotent Xuejian Fang	
	fix(release): set HEAD_SHA on main and make gh release creation idempotent xfan0282	
	Merge pull request #59 from SOFT2412-COMP9412-2025s2/chore/ci-semver-release Xuejian Fang	chore/ci-semver...
	chore(release): disable test-mode so releases run only on main xfan0282	
	fix(release): drive gh via GH_HOST/GH_TOKEN (remove --hostname), write CHANGELOG.txt, and guard release idempotently xfan0282	
	ci(release): switch to GitHub CLI (gh) for tagging and releases; set release author xfan0282	
	ci(release): set git committer identity before tagging (fix test-mode release) xfan0282	
	ci(release): temporarily enable real tag+release on non-main to validate flow xfan0282	
	ci(release): auto tag & GitHub release on main (API-first + title fallback) xfan0282	
	ci: make API retry POSIX-safe and normalize tag prefix (V/V) xfan0282	
	ci: fix dry-run auth for fetching tags (use HTTPS PAT) xfan0282	
	ci: add API-first next-version calculator with title fallback (dry-run on non-main) xfan0282	
	chore(cli): add VERSION_MAJOR xfan0282	
	Merge pull request #58 from SOFT2412-COMP9412-2025s2/fix/gradle-cache-perms Xuejian Fang	
	fix(cli): correct JaCoCo HTML path and configure coverage source directories xfan0282	
	fix(cli): redirect Gradle user home to workspace to avoid EPERM on -/gradle; disable implicit checkout xfan0282	
	Merge pull request #55 from SOFT2412-COMP9412-2025s2/ci/coverage-and-html-publish Boning Zhao	
	ci: switch to Coverage plugin and publish JaCoCo HTML xfan0282	
	Merge pull request #54 from SOFT2412-COMP9412-2025s2/test/increase-coverage-rate Xuanhao Yu	
	set code style xuanhao	
	Merge pull request #53 from SOFT2412-COMP9412-2025s2/test/increase-coverage-rate Xuanhao Yu	
	add file scroll repository edge test, instruction coverage rate increase to 85% xuanhao	
	add file user repository duplicate test, branch coverage rate increase to 60% xuanhao	
	add commanddispatchertest, instruction, instruction coverage increase to 84%, branch coverage rate increase to 59% xuanhao	
	Merge pull request #52 from SOFT2412-COMP9412-2025s2/fix/cli-usage-help-text Boning Zhao	
	fix(docs): update test report paths in README bzh8676	
	delete(data): remove obsolete scrolls.tsv file bzh8676	
	fix(cli): align usage text with commands bzh8676	
	Merge pull request #51 from SOFT2412-COMP9412-2025s2/chore/gitignore-data Xuanhao Yu	
	add "data" folder in gitignore xuanhao	
	Merge pull request #50 from SOFT2412-COMP9412-2025s2/feature/cli-session Boning Zhao	
	feat(cli): Implement session persistence in Login and Logout commands bzh8676	
	Merge pull request #49 from SOFT2412-COMP9412-2025s2/feature/interactive-shell Boning Zhao	
	chore: add gradle properties to configure console output bzh8676	
	Add interactive shell mode for VSAS CLI bzh8676	
	Merge pull request #48 from SOFT2412-COMP9412-2025s2/feature/us-a1-Register-account Xuejian Fang	
	test(cli): cover register success/duplicate/missing and no-plaintext (A3-T28-G03#15) xfan0282	
	feat(cli): enforce unique idKey in register (A3-T28-G03#14) xfan0282	
	feat(cli): Implement vsas register with salted hashing via repository (A3-T28-G03#13) xfan0282	
	feat(security): add salted SHA-256 password hasher with tests (A3-T28-G03#10) xfan0282	
	feat(repo): implement TSV-backed FileUserRepository with minimal tests (A3-T28-G03#9) xfan0282	
	feat(model): add User.createdAt and backward-compatible constructor (A3-T28-G03#8) xfan0282	
	Merge pull request #47 from SOFT2412-COMP9412-2025s2/feat/scroll-list-command Cheng Zhiwei Cheng	
	feat(scroll): add list command with TSV-backed repositor Zche0826	
	Merge pull request #46 from SOFT2412-COMP9412-2025s2/fix/formatting-registercommand-authhashingtest Xuejian Fang	
	chore(format): re-run googleJavaFormat for RegisterCommand.java and AuthHashingIntegrationTest.java xfan0282	
	Merge pull request #45 from SOFT2412-COMP9412-2025s2/feature/us-a6-hashed-passwords Xuejian Fang	
	test(auth): add salt-variance test for same password (A3-T28-G03#41) xfan0282	
	feat(auth): hash password on registration and persist salt (A3-T28-G03#39) xfan0282	
	feat(auth): verify hashed passwords during login (A3-T28-G03#40) xfan0282	
	feat(security): add salted password hashing utility (A3-T28-G03#38) xfan0282	
	Merge pull request #44 from SOFT2412-COMP9412-2025s2/feat/cli-implement-upload-command Cheng Zhiwei Cheng	
	Merge branch 'main' into feat/cli-implement-upload-command Cheng Zhiwei Cheng	
	style(upload): use google-java-format format UploadCommand.java Zche0826	
	feat(cli): Implement UploadCommand with file copy and metadata persistence Zche0826	
	Merge pull request #42 from SOFT2412-COMP9412-2025s2/feat/scroll/cli-upload-unique-id Zhiwei Cheng	
	feat(cli): update CommandDispatcher to return success on usage help and enhance FileScrollRepository and SessionService for scroll management Zche0826	
	Merge pull request #12 from SOFT2412-COMP9412-2025s2/feature/sprint1-cli-skeleton Boning Zhao	
	chore(cli): add runnable CLI skeleton and package structure bzh8676	
	Merge pull request #6 from SOFT2412-COMP9412-2025s2/fix/ci-gradlew-and-format-check Xuejian Fang	
	ci: re-enable googleJavaFormat check with non-blocking behavior xfan0282	
	fix(cli): make gradlew executable for CI and local usage xfan0282	
	Merge pull request #5 from SOFT2412-COMP9412-2025s2/chore/configure-project-tooling Xuejian Fang	
	chore(build): configure initial testing and formatting tools bzh8676	
	Merge branch 'main' of https://github.com/sydney.edu.au/SOFT2412-COMP9412-2025s2/CC28-Team03-AZA3 bzh8676	
	Merge pull request #4 from SOFT2412-COMP9412-2025s2/fix/ci-jenkinsfile-fix Xuejian Fang	
	fix(cli): remove unsupported ansiColor option; ensure gradlew exec & skip google-java-format xfan0282	
	fix(ci): ensure gradlew executable in CI, skip verifyGoogleJavaFormat, publish JUnit and archive JARs; safer deploy xfan0282	
	Merge pull request #3 from SOFT2412-COMP9412-2025s2/chore/bootstrap-with-gradle-init Boning Zhao	
	chore: initialize project via Gradle init and add CI bzh8676	
	Revert "chore: add initial build.gradle file for Java application setup" bzh8676	
	chore: add initial build.gradle file for Java application setup bzh8676	
	Merge pull request #2 from SOFT2412-COMP9412-2025s2/docs/readme-and-gitignore Boning Zhao	
	docs: add README with run/test instructions and project .gitignore bzh8676	
	Merge pull request #1 from SOFT2412-COMP9412-2025s2/feature/add-jenkinsfile Xuejian Fang	
	ci(jenkins): add Jenkinsfile for Gradle multi-branch pipeline xfan0282	
	Initial commit Xuejian Fang	

SOFT2412-COMP9412-2025s2 / A3-T28-G03

Type // to search

<> Code

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Settings

The maintenance has been rescheduled to Thursday, 13 November 2025, from 5:00 PM to 10:00 PM. The service will be unavailable during this period.

A3-T28-G03

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About

xfan0282 Merge pull request #107 from SOFT2412-COMP9412-2025s2/do... d9bb459 · 2 days ago 182 Commits

app	feat: enhance registration command to support role speci...	2 days ago
gradle	chore: initialize project via Gradle init and add CI	3 weeks ago
.gitattributes	chore: initialize project via Gradle init and add CI	3 weeks ago
.gitignore	add "data" folder in gitignore	2 weeks ago
Jenkinsfile	fix: rename vavs to vsas	2 days ago
LICENSE	Initial commit	3 weeks ago
README.md	docs: update new jar name	2 days ago
VERSION_MAJOR	chore: bump VERSION_MAJOR to 3	5 days ago
gradle.properties	chore: add gradle.properties to configure console output	2 weeks ago
gradlew	fix(ci): make gradlew executable for CI and local usage	2 weeks ago
gradlew.bat	chore: initialize project via Gradle init and add CI	3 weeks ago
settings.gradle	chore: initialize project via Gradle init and add CI	3 weeks ago

README

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Readme

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V3.8.3 Latest 2 days ago

+ 46 releases

Contributors 4

xfan0282 Xuejian Fang

bzha8676 Borling Zhao

xuyu8020 Xuanhao Yu

zche0826 Cheng Zhiwei Cheng

US-UX1 in-app help for existing commands #65

6 tasks done

bzha8676 opened this issue 2 weeks ago · 0 comments · Fixed by #87

bzha8676 commented 2 weeks ago · edited

Member

Story

As a CLI User, I want in-app guidance in the REPL so I can discover and use commands without having to remember flags.

Acceptance Criteria

- When the REPL starts, show a one-line welcome and Type 'help' or 'help <command>'.
- help lists all available commands with short descriptions; help <command> prints usage, required/optional flags, and one example.
- Command names and help are generated from a single command registry/metadata used by both the dispatcher and help (no hardcoded lists).
- Standard exit codes across all commands: 0=success, 1=validation/permission, 2=usage, 3=IO.

Tasks

- Create a CommandRegistry that centralises command metadata (name, description, usage, flags, example, exit codes) and exposes lookup/iteration APIs for the dispatcher and help flow.
- Refactor CommandDispatcher to consult the registry instead of a switch, adding support for grouped commands (e.g. scroll, profile) and keeping the Usage output derived from the registry.
- Print a one-line welcome banner when the REPL starts (App.runInteractive), instructing users to type help or help <command>.
- Implement HelpCommand that renders the full command list and per-command usage from the registry, and register it alongside existing commands.
- Audit existing commands to align with the standard exit codes (0 success, 1 validation/permission, 2 usage, 3 IO) and update affected tests.
- Add tests covering registry wiring, help output for global and specific commands, and the REPL banner.

Assignees

bzha8676

Labels

None yet

Projects

A3-T28-G03

Status: Done

Milestone

No milestone

Development

Successfully merging a pull request may close this issue.

Feature/us ux1 help registry

SOFT2412-COMP9412-2025s2/A3-T28-G03

Notifications

Unsubscribe

1 participant

Clear current search query, filters, and sorts

☐	0 Open ✓ 56 Closed	Author	Label	Projects	Milestones	Reviews	Assignee	Sort
☐	docs: update new jar name ✓ #107 by xfan0282 was merged 2 days ago · Approved							
☐	Docs/improve readme ✓ #106 by xfan0282 was merged 2 days ago · Approved					1		
☐	Chore/release rename jar to vavs ✓ #105 by xfan0282 was merged 2 days ago · Approved							
☐	chore(release): upload fat jar as vavs-<version>.jar • #104 by xfan0282 was closed 2 days ago · Review required							
☐	feat: enhance registration command to support role specification and ... ✓ #103 by zche0826 was merged 2 days ago · Approved							
☐	Feature/us b2 admin add users ✓ #102 by xfan0282 was merged 3 days ago · Approved					1		
☐	Feature/us b2 admin add users ✗ #101 by xfan0282 was closed 3 days ago · Review required							
☐	Feature/us b1 admin users list ✓ #100 by xfan0282 was merged 3 days ago · Approved					1		
☐	US-B5 Feat/admin stats ✓ #99 by xuyu8020 was merged 3 days ago · Approved					1		
☐	feat: implement admin user role update command and repository support ✓ #98 by zche0826 was merged 3 days ago · Approved					1		
☐	feat: add admin user deletion command and repository support ✓ #97 by zche0826 was merged 3 days ago · Approved					1		
☐	Feat/us bm1 bookmark cli ✓ #96 by bzha8676 was merged 3 days ago · Approved					1		
☐	chore: bump VERSION_MAJOR to 3 ✓ #88 by xfan0282 was merged 5 days ago · Approved							
☐	Feature/us ux1 help registry ✓ #87 by bzha8676 was merged 5 days ago · Approved					1		
☐	Chore/remove legacy list output ✓ #86 by bzha8676 was merged last week · Approved							
☐	Fix/login single prompt and usage ✓ #85 by xfan0282 was merged last week · Approved					1		
☐	Feat/us a3 update profile ✓ #84 by bzha8676 was merged last week · Approved					1		
☐	feat: track scroll upload/download counts - US-E1 ✓ #82 by zche0826 was merged last week · Approved					1		
☐	Feat/us d4 preview scroll ✓ #81 by zche0826 was merged last week · Approved					1		1
☐	Feat/d3 search filters ✓ #80 by xfan0282 was merged last week · Approved					1		
☐	Feat/g1 secure password prompt ✓ #79 by xfan0282 was merged last week · Approved					1		
☐	US-C2 Feat/scroll update ✓ #78 by xuyu8020 was merged last week · Approved					1		
☐	Feat/us d4 preview scroll ✗ invalid wontfix #77 by bzha8676 was closed last week · Review required							
☐	fix(release): anchor feat detection to PR source branch and support s... ✓ #76 by xfan0282 was merged last week · Approved							1
☐	Fix/release feat detection ✓ #75 by xfan0282 was merged last week · Review required							

ProTip! Notify someone on an issue with a mention, like: @xfan0282.

3.2 CI/CD Tools and Practices

3.2.1 CI Pipeline (Jenkins)

Unchanged from Sprint 1/2: Jenkins Multibranch Pipeline runs a single repo-defined Jenkinsfile: Checkout → Sanity → Format (verify) → Build (Gradle) → Test (JUnit + JaCoCo) → Archive → Deploy (placeholder on main only) → Release (main only). Recent builds remain green with tests passing and artifacts archived; turnaround time continues to be fast, providing quick feedback.

3.2.2 Status Visibility in GitHub

Unchanged: Every commit/PR surfaces a Jenkins status check in GitHub. Failing checks block merges; passing checks show a green tick and allow review/merge per branch protection rules.

3.2.3 Triggers and Branch Discovery

Unchanged: Multibranch auto-discovers feature/* branches and PRs, building both PR head and the “head merged into target” simulation. This validates changes in the context of the protected target branch.

3.2.4 Build, Test, and Coverage

Unchanged: Style gate (google-java-format) runs early and fails fast; Gradle build produces jars; JUnit + JaCoCo generate XML/HTML with Jenkins publishing/archiving reports for auditability.

3.2.5 Automated Versioning & Release

Unchanged: Jenkins computes the semantic version, tags the repo, creates a GitHub Release, and uploads the runnable artifact on main-branch merges only. Versioning rules (major via VERSION_MAJOR, minor/patch via commit type) and tag prefix detection remain as in Sprint 2. The release artifact continues to be published as a versioned jar (vsas-<version>.jar) with fingerprints and traceability to the exact CI run.

3.2.6 CD Status

Unchanged: Scope remains Continuous Integration + delivery to GitHub Releases. No environment deployment yet; test/pre-prod CD work is still planned for a future iteration.

3.2.7 Reliability & Hosting

Unchanged: Jenkins remains containerised on a 24/7 host with stable DNS; no outages or configuration changes were required this sprint. Overall CI/CD flow is stable and operating as in Sprint 2, with no new functionality introduced this week.

jenkins.kscii.tech

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Add description

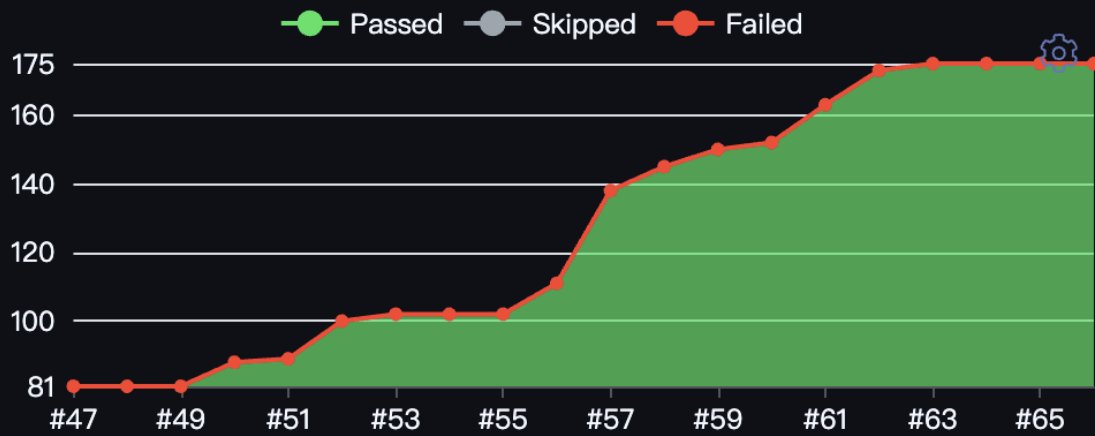
All

S	W	Name ↓	Last Success	Last Failure	Last Duration	Coverage	F
		A3-T28-G03	11 days log	N/A	4.4 sec		

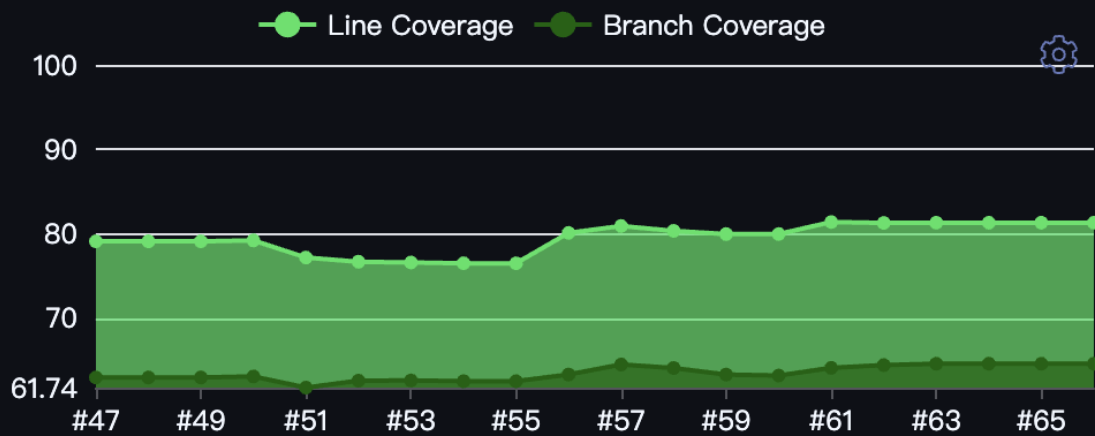
Icon: S M L

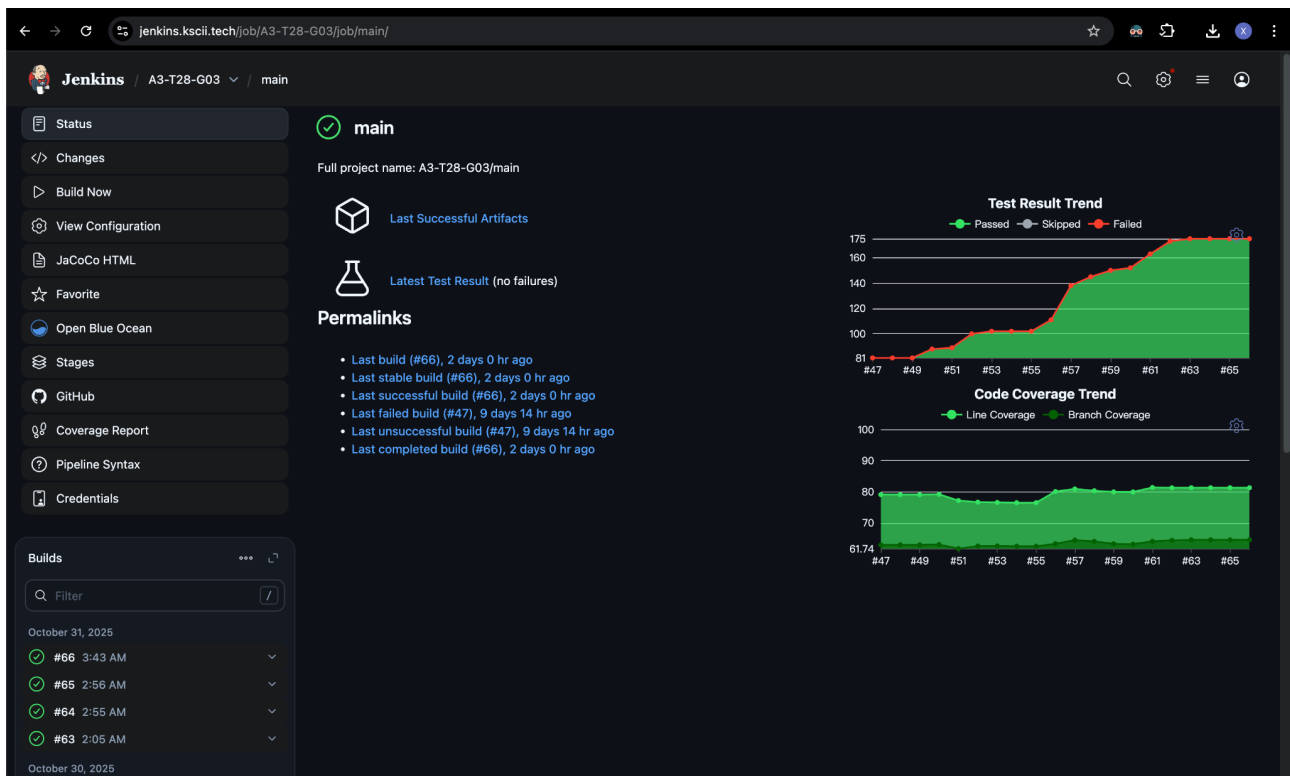
...

Test Result Trend



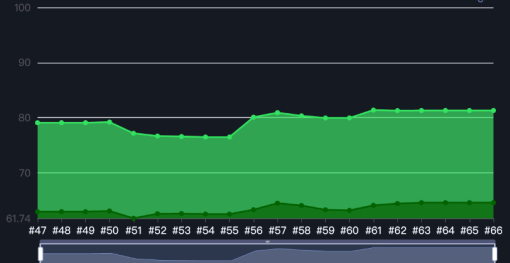
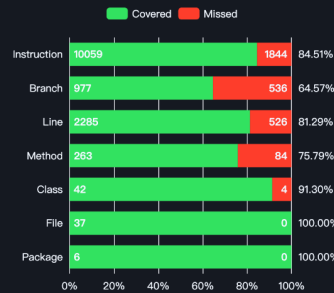
Code Coverage Trend





- ## Coverage Report

Files



Timings

		Primary task	Including subtasks
In queue	Waiting	7.6 sec	7.6 sec
	Blocked	0 ms	0 ms
	Buildable	0 ms	10 ms
	Total	7.6 sec	7.6 sec
Building		3 min 29 sec	3 min 28 sec
Scheduled to completion			3 min 37 sec
Number of subtasks			1
Average executor utilization			1.0

```

14:46:41     "author_association": "MEMBER",
14:46:41     "auto_merge": null,
14:46:41     "active_lock_reason": null
14:46:41   }
14:46:41 ]
14:46:41 + SRC_BRANCH=
14:46:41 + [ -n ]
14:46:41 + sleep 4
14:46:45 + git log -1 --pretty=%s
14:46:45 + MERGE_SUBJ=Merge pull request #107 from SOFT2412-COMP9412-2025s2/docs/improve-readme
14:46:45 + + printf %sgrep -Eiq
14:46:45 ^feat|feature)(/|-)
14:46:45 + + printf %sgrep -Eiq Merge pull request #107 from SOFT2412-COMP9412-2025s2/docs/improve-readme
14:46:45 ^Merge pull request #[0-9]+ from [^ ]+/(feat|feature)(/|-)
14:46:45 + + printf %sgrep -Eiq Merge pull request #107 from SOFT2412-COMP9412-2025s2/docs/improve-readme
14:46:45 ^feat(((?!/)|/|-))
14:46:45 + MINOR=8
14:46:45 + PATCH=3
14:46:45 + NEXT_VERSION=V3.8.3
14:46:45 + + echo V3.8.3tee next-version.txt
14:46:45
14:46:45 V3.8.3
14:46:45 + git rev-parse -q --verify refs/tags/V3.8.3
14:46:45 + git tag -a V3.8.3 -m Release V3.8.3 (automated by Jenkins)
14:46:45 + git push origin V3.8.3
14:46:46 To https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03.git
14:46:46 * [new tag]          V3.8.3 -> V3.8.3
14:46:46 + [ -n V3.8.2..HEAD ]
14:46:46 + echo Changes since V3.8.2:
14:46:46 + echo
14:46:46 + git log --no-merges --pretty=* %h %s (%an) V3.8.2..HEAD
14:46:46 + export GH_TOKEN=***
14:46:46 + export GH_HOST=github.sydney.edu.au
14:46:46 + gh release view V3.8.3 -R SOFT2412-COMP9412-2025s2/A3-T28-G03
14:46:46 + gh release create V3.8.3 -R SOFT2412-COMP9412-2025s2/A3-T28-G03 --target d9bb45932827ea23c5ef32e9557776e61a4d7589 --title V3.8.3 --notes-file CHANGELOG.txt
14:46:46 https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/releases/tag/V3.8.3
14:46:46 + echo Release V3.8.3 created via gh.
14:46:46 Release V3.8.3 created via gh.
14:46:46 + JAR_GLOB=app/build/libs/*-all.jar
14:46:46 + head -n1
14:46:46 + ls app/build/libs/app-all.jar
14:46:46 + FIRST_JAR=app/build/libs/app-all.jar
14:46:46 + VER_NUM=3.8.3
14:46:46 + VER_NUM=3.8.3
14:46:46 + OUT_NAME=vsas-3.8.3.jar
14:46:46 + cp app/build/libs/app-all.jar vsas-3.8.3.jar
14:46:47 + echo Uploading vsas-3.8.3.jar
14:46:47 Uploading vsas-3.8.3.jar
14:46:47 + gh release upload V3.8.3 vsas-3.8.3.jar -R SOFT2412-COMP9412-2025s2/A3-T28-G03 --clobber
[Pipeline] archiveArtifacts
14:46:47 Archiving artifacts
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
14:46:49 Pipeline completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // timestamps
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline

GitHub has been notified of this commit's build result

Finished: SUCCESS

```

3.3 Code Quality

3.3.1 Tooling and Linting

Unchanged from Sprint 2: google-java-format (Gradle plugin) standardises formatting. Developers run `./gradlew googleJavaFormat` locally; Jenkins runs `verifyGoogleJavaFormat` on every build. Tests use Gradle + JUnit 5, and JaCoCo executes on each CI run (`jacocoTestReport`). GitHub continues to surface a green check on commits/PRs when CI passes.

3.3.2 Tests and Coverage

Approach and locations unchanged. Unit tests remain in place across CLI commands, repositories, session service, and password hashing; JaCoCo HTML/XML reports are published on every build. This sprint reuses the established baseline without process or threshold changes.

3.3.3 Code Reviews

No changes to practice. Protected-branch rules continue to enforce “CI must be green” and at least one reviewer approval before merge. PR conventions (small, focused changes tied to task IDs) remain the same.

3.3.4 Coding Standards

No changes. We continue to follow Google Java Style via google-java-format, keep public contracts stable, and maintain incremental, well-scoped commits that reference task IDs with conventional prefixes. Overall code-quality process is stable and unchanged this sprint.

```

kscii@mac1 A3-T28-G03 % tree app/src
app/src
├── main
│   ├── java
│   │   ├── org
│   │   │   ├── soft2412
│   │   │   │   ├── vsas
│   │   │   │   │   ├── App.java
│   │   │   │   │   ├── cli
│   │   │   │   │   │   ├── AdminStatsCommand.java
│   │   │   │   │   │   ├── AdminUsersAddCommand.java
│   │   │   │   │   │   ├── AdminUsersDeleteCommand.java
│   │   │   │   │   │   ├── AdminUsersListCommand.java
│   │   │   │   │   │   ├── AdminUsersRoleCommand.java
│   │   │   │   │   │   ├── BookmarkAddCommand.java
│   │   │   │   │   │   ├── BookmarkListCommand.java
│   │   │   │   │   │   ├── BookmarkRemoveCommand.java
│   │   │   │   │   │   ├── Command.java
│   │   │   │   │   │   ├── CommandDispatcher.java
│   │   │   │   │   │   ├── CommandHelp.java
│   │   │   │   │   │   ├── CommandRegistration.java
│   │   │   │   │   │   ├── CommandRegistry.java
│   │   │   │   │   │   ├── CommandResolution.java
│   │   │   │   │   │   ├── DownloadCommand.java
│   │   │   │   │   │   ├── HelpCommand.java
│   │   │   │   │   │   ├── ListCommand.java
│   │   │   │   │   │   ├── LoginCommand.java
│   │   │   │   │   │   ├── LogoutCommand.java
│   │   │   │   │   │   ├── PasswordPrompt.java
│   │   │   │   │   │   ├── PreviewCommand.java
│   │   │   │   │   │   ├── ProfileUpdateCommand.java
│   │   │   │   │   │   ├── RegisterCommand.java
│   │   │   │   │   │   ├── ScrollCommand.java
│   │   │   │   │   │   ├── ScrollDeleteSubcommand.java
│   │   │   │   │   │   ├── ScrollUpdateSubcommand.java
│   │   │   │   │   │   ├── UploadCommand.java
│   │   │   │   │   │   ├── WhoAmICommand.java
│   │   │   │   │   ├── model
│   │   │   │   │   │   ├── Bookmark.java
│   │   │   │   │   │   ├── Scroll.java
│   │   │   │   │   │   ├── ScrollUsage.java
│   │   │   │   │   │   ├── User.java
│   │   │   │   │   ├── repo
│   │   │   │   │   │   ├── BookmarkRepository.java
│   │   │   │   │   │   ├── FileBookmarkRepository.java
│   │   │   │   │   │   ├── FileScrollRepository.java
│   │   │   │   │   │   ├── FileUserRepository.java
│   │   │   │   │   │   ├── ScrollRepository.java
│   │   │   │   │   │   ├── UserRepository.java
│   │   │   │   │   ├── security
│   │   │   │   │   │   ├── PasswordHasher.java
│   │   │   │   │   ├── service
│   │   │   │   │   │   ├── SessionService.java
│   │   │   │   │   ├── test
│   │   │   │   │   │   ├── java
│   │   │   │   │   │   │   ├── org
│   │   │   │   │   │   │   │   ├── soft2412
│   │   │   │   │   │   │   │   │   ├── vsas
│   │   │   │   │   │   │   │   │   ├── AppInteractiveTest.java
│   │   │   │   │   │   │   │   │   ├── AppTest.java
│   │   │   │   │   │   │   │   │   ├── cli
│   │   │   │   │   │   │   │   │   │   ├── AdminStatsCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── AdminUsersAddCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── AdminUsersDeleteCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── AdminUsersListCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── AdminUsersRoleCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── AuthHashingIntegrationTest.java
│   │   │   │   │   │   │   │   │   │   ├── BookmarkCommandsTest.java
│   │   │   │   │   │   │   │   │   │   ├── CommandDispatcherTest.java
│   │   │   │   │   │   │   │   │   │   ├── CommandRegistryTest.java
│   │   │   │   │   │   │   │   │   │   ├── DownloadCommandCountTest.java
│   │   │   │   │   │   │   │   │   │   ├── DownloadCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── HelpCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── LoginCommandInteractiveTest.java
│   │   │   │   │   │   │   │   │   │   ├── LoginCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── LogoutCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── PasswordPromptTest.java
│   │   │   │   │   │   │   │   │   │   ├── PreviewCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── ProfileUpdateCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── RegisterCommandInteractiveTest.java
│   │   │   │   │   │   │   │   │   │   ├── RegisterCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── RegisterCommandUniqueIdKeyTest.java
│   │   │   │   │   │   │   │   │   │   ├── ScrollCommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── ScrollDeleteSubcommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── ScrollUpdateSubcommandTest.java
│   │   │   │   │   │   │   │   │   │   ├── ScrollUpdateUploadCountTest.java
│   │   │   │   │   │   │   │   │   ├── FileScrollRepositoryTest.java
│   │   │   │   │   │   │   │   │   ├── ListCommandFiltersAndFormattingTest.java
│   │   │   │   │   │   │   │   │   ├── ListCommandTest.java
│   │   │   │   │   │   │   │   │   ├── model
│   │   │   │   │   │   │   │   │   │   ├── UserTest.java
│   │   │   │   │   │   │   │   │   ├── repo
│   │   │   │   │   │   │   │   │   │   ├── FileBookmarkRepositoryTest.java
│   │   │   │   │   │   │   │   │   │   ├── FileScrollRepositoryEdgeTest.java
│   │   │   │   │   │   │   │   │   │   ├── FileUserRepositoryDuplicateTest.java
│   │   │   │   │   │   │   │   │   │   ├── FileUserRepositoryTest.java
│   │   │   │   │   │   │   │   │   ├── security
│   │   │   │   │   │   │   │   │   │   ├── PasswordHasherTest.java
│   │   │   │   │   │   │   │   │   ├── SessionServiceTest.java
│   │   │   │   │   │   │   │   │   ├── UploadCommandTest.java
│   │   │   │   │   │   │   │   │   ├── WhoAmICommandTest.java

```

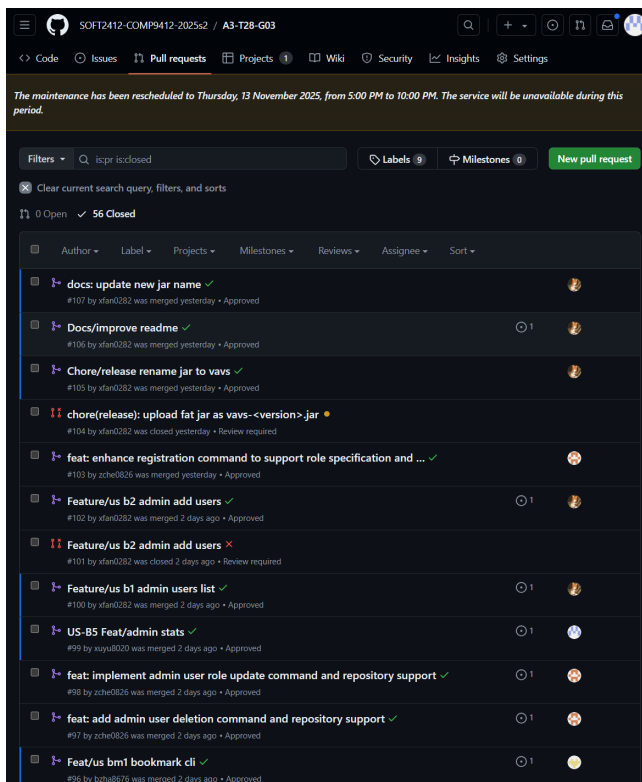
20 directories, 80 files

4. Application Development

4.1 Collaborative Coding

In Sprint 3, the team continued to use the feature-branch model to complete the final integration of VSAS. Each member developed new features or fixed bugs on independent branches, named `feat/<feature>` or `fix/<issue>`, and conducted peer reviews and CI checks through Pull Requests before merging. All branches were regularly synchronised with the main branch to ensure code consistency.

This sprint focused on improving the previous modules and adding bookmark functionality, enhancing the administrator report, and improving the CLI user experience. All new features were automatically tested through Gradle, with a coverage rate exceeding 80%. All PRs required approval from at least one reviewer before merging, and the commit messages and formats followed the Google Java Style standard.



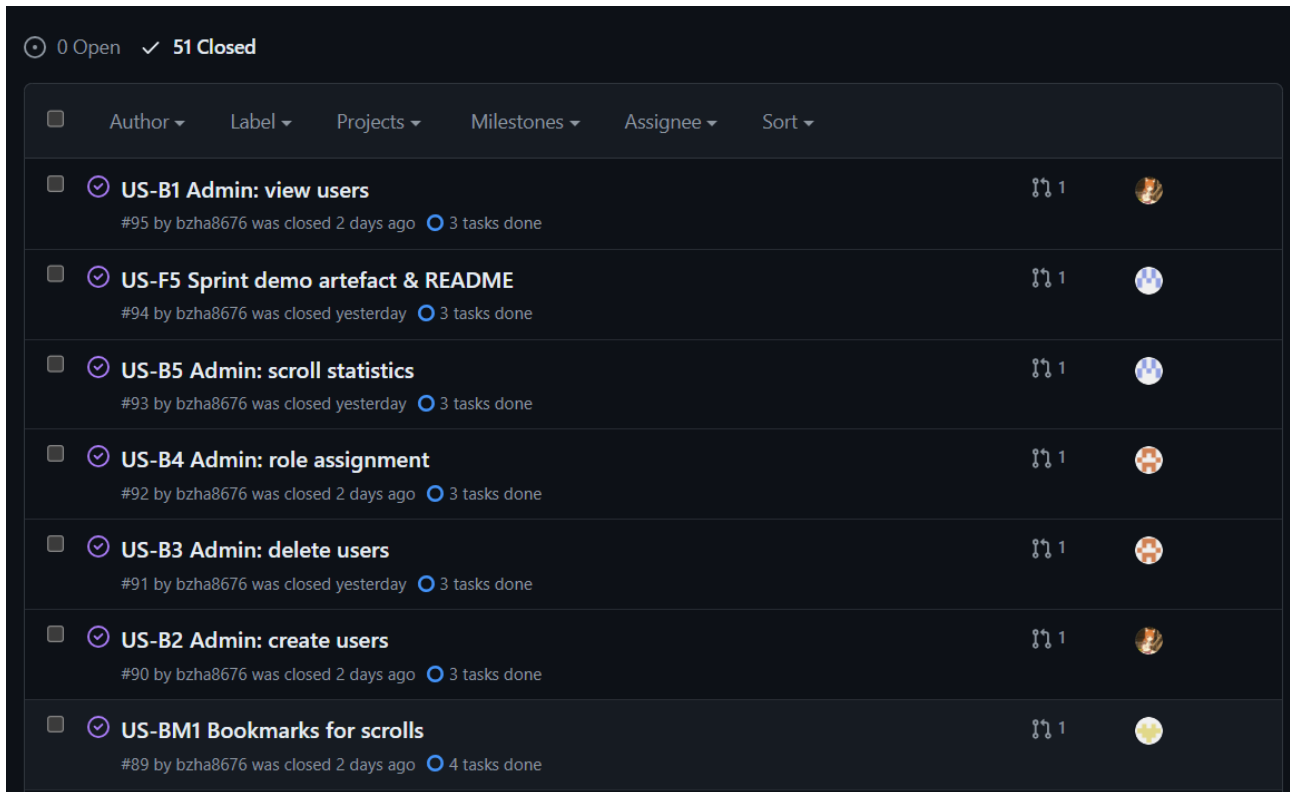
4.2 Individual and Group Contributions

Xuejian Fang (Scrum Master) — Implemented US-B1 and US-B2 (AdminUsersListCommand, AdminUsersAddCommand) to enable admins to view and create users through CLI. He also enhanced GitHub Release automation to auto-attach version tags for each release.

Zhiwei Cheng (Developer) — Completed US-B3 and US-B4 (AdminUsersDeleteCommand, AdminUsersRoleCommand), adding role-assignment and safe-deletion mechanisms with proper error handling.

Xuanhao Yu (Developer) — Implemented US-B5 (Admin Statistics) and US-F5 (Sprint Demo Artefact & README). He packaged the final CLI JAR (vsas-cli.jar) and verified it could run end-to-end without an IDE.

Boning Zhao (Product Owner) — Completed US-BM1 (Bookmarks for Scrolls) and developed an enhanced Help command system, allowing users to type help to see all available commands, and help <command> to view required and optional flags. He also authored and standardised User Story documentation on GitHub.



The screenshot shows a GitHub Issues page with a dark theme. At the top, it indicates '0 Open' and '51 Closed' issues. Below this is a filter bar with dropdown menus for 'Author', 'Label', 'Projects', 'Milestones', 'Assignee', and 'Sort'. The main content is a list of seven closed issues, each with a checkbox, a checkmark icon, a title, a description, a status indicator (3 tasks done), a link icon, and an assignee's profile picture.

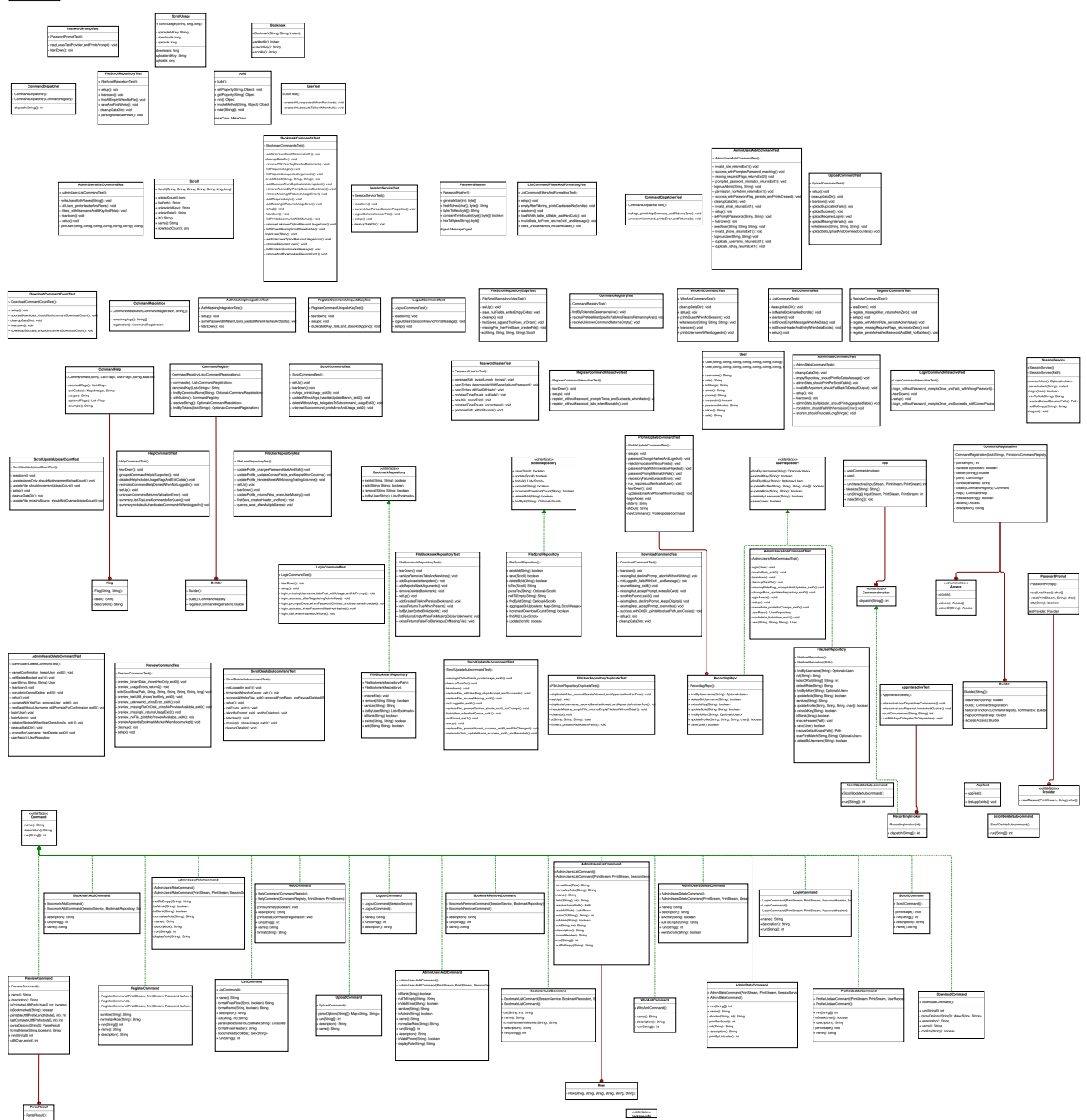
☐	Author ▾	Label ▾	Projects ▾	Milestones ▾	Assignee ▾	Sort ▾
☐	✓	US-B1 Admin: view users	#95 by bzha8676 was closed 2 days ago	3 tasks done	1	
☐	✓	US-F5 Sprint demo artefact & README	#94 by bzha8676 was closed yesterday	3 tasks done	1	
☐	✓	US-B5 Admin: scroll statistics	#93 by bzha8676 was closed yesterday	3 tasks done	1	
☐	✓	US-B4 Admin: role assignment	#92 by bzha8676 was closed 2 days ago	3 tasks done	1	
☐	✓	US-B3 Admin: delete users	#91 by bzha8676 was closed yesterday	3 tasks done	1	
☐	✓	US-B2 Admin: create users	#90 by bzha8676 was closed 2 days ago	3 tasks done	1	
☐	✓	US-BM1 Bookmarks for scrolls	#89 by bzha8676 was closed 2 days ago	4 tasks done	1	

4.3 CI (Continuous Integration) Tools and Structure Management

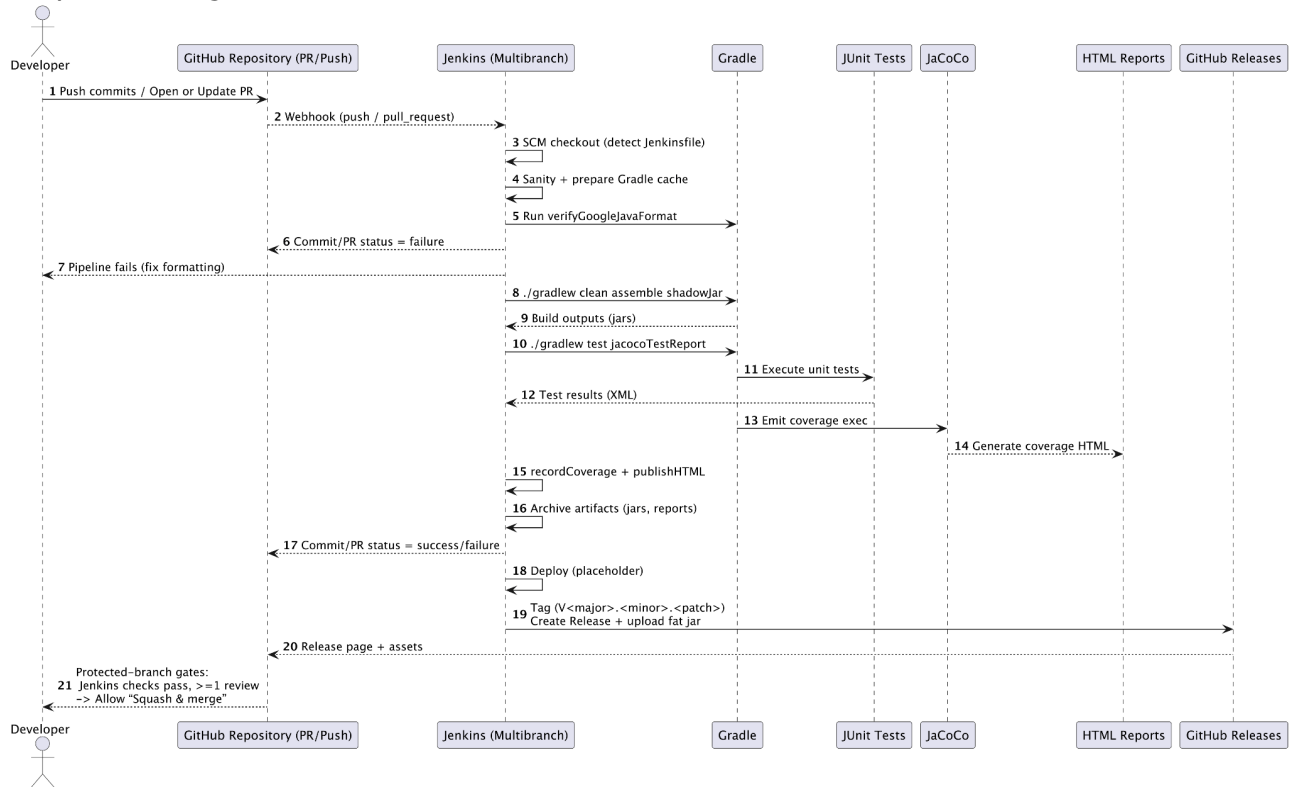
The Jenkins pipeline can automatically build, test and release vsas-cli.jar whenever a PR is triggered.

The project maintains a modular architecture and new modules such as Help, Admin and Bookmark have been added.

AgentJar
AgentJar()
RESOURCE: String
+ extractFile(): void
+ setDownloadTimeout(): void
+ extractToTempLocation(): File
resourceAbsolutePath: InputStream
RESOURCE: URL



Sequence Diagram:



4.4 Functional and Non-functional Aspects

Functional Requirements Achieved :

Admins can view, create, delete users and assign roles (US-B1 ~ B4).

Admins can view scroll statistics (US-B5).

Users can add and remove bookmarks (US-BM1).

Enhanced Help Command (US-BM1 extension): help lists all available commands, help <command> displays required and optional flags.

Non-Functional Improvements :

Reliability — Jenkins CI/CD ensures stable releases.

Usability — Improved Help system and CLI feedback.

```
vsas> help
Available commands:
register - Register a new user
login - Log in with username and password
logout - Log out of the current session
whoami - Show the current authenticated user or guest
list - List scroll metadata with optional filters
upload - Upload a scroll binary and metadata
download - Download a scroll to the local filesystem
bookmark - Manage personal scroll bookmarks
scroll - Manage scroll lifecycle commands
preview - Preview metadata and a snippet of a scroll
profile - Manage profile details
help - Show help for commands
```

```
> Task :app:run
Welcome to VSAS CLI. Type 'help' or 'help <command>' for assistance.
vsas> login --username test1 --password 00000000
Login success
vsas> whoami
test1 (role=ADMIN)
vsas> admin users list
username      email                phone      idKey      role      createdAt
test1         123@321.com          123123123  1          admin    2025-10-31T04:06:35.
test3         123@321.com          123456789  3          user     2025-11-01T10:51:20.
vsas> admin users role --username test3 --role admin
Role updated: test3 -> admin
vsas> admin users list
username      email                phone      idKey      role      createdAt
test1         123@321.com          123123123  1          admin    2025-10-31T04:06:35.
test2         123@321.com          123123123  2          user     2025-10-31T04:06:41.
test3         123@321.com          123456789  3          admin    2025-11-01T10:51:20.
vsas> admin users delete --username test3
username=test3, idKey=3, role=ADMIN
Delete user test3? This cannot be undone. [y/N] y
Deleted user test3.
vsas> admin users list
username      email                phone      idKey      role      createdAt
test1         123@321.com          123123123  1          admin    2025-10-31T04:06:35.
test2         123@321.com          123123123  2          user     2025-10-31T04:06:41.
```

```

vsas> admin stats
ID          NAME          UPLOADER          UPLOADS          DOWNLOADS
-----
1          readme          1                  1                  0
2          gitignore        1                  1                  0
vsas> admin stats --by uploader
UPLOADER          UPLOADS          DOWNLOADS
-----
1                  2                  0
2                  1                  0
vsas> scroll update --id 1 --name "ReadMe" --file "C:\Users\32234\code\Java\2412-2025\A3-T28-G03\README.md"
Replace existing file? [y/n] y
Updated: 1
vsas> admin stats
ID          NAME          UPLOADER          UPLOADS          DOWNLOADS
-----
1          ReadMe          1                  2                  0
2          gitignore        1                  1                  0
vsas> download --id 1 --out C:\Users\32234\code\Java\2412-2025
C:\Users\32234\code\Java\2412-2025\1. bin
vsas> admin stats
ID          NAME          UPLOADER          UPLOADS          DOWNLOADS
-----
1          ReadMe          1                  2                  1
2          gitignore        1                  1                  0
vsas> admin stats --by uploader
UPLOADER          UPLOADS          DOWNLOADS
-----
1                  3                  1

```

```

vsas> bookmark list
No bookmarks.
vsas> bookmark add --id 1
Bookmarked 1
vsas> bookmark list
sid          name          uploaderIdKey    addedAt
1          ReadMe [BK]    1                2025-11-01T10:56:21.25611
vsas> bookmark add --id 2
Bookmarked 2
vsas> bookmark list
sid          name          uploaderIdKey    addedAt
1          ReadMe [BK]    1                2025-11-01T10:56:21.25611
2          gitignore [BK] 1                2025-11-01T10:56:31.13360
vsas> bookmark remove --id 1
Remove bookmark 1? [y/N] y
Removed bookmark 1
vsas> bookmark list
sid          name          uploaderIdKey    addedAt
2          gitignore [BK] 1                2025-11-01T10:56:31.13360
vsas> bookmark remove --id 2
Remove bookmark 2? [y/N] y
Removed bookmark 2
vsas> bookmark list
No bookmarks.

```

5. Group Contribution

5.1 Team Organisation and Roles

The team followed the Scrum methodology to maintain a self-organised and iterative development process.

Roles were clearly defined in accordance with the Scrum structure:

Product Owner – Boning Zhao

Responsible for defining user requirements, prioritising backlog items, and ensuring that implemented features aligned with stakeholder expectations.

In addition to managing the product backlog, he contributed to the system's architecture by creating the initial Gradle-based scaffold and standardising the project's directory hierarchy.

Scrum Master – Xuejian Fang

Oversaw sprint planning, progress tracking, and facilitated team communication to ensure adherence to Scrum principles.

From a technical perspective, he implemented the CI/CD infrastructure, configuring the Jenkins multibranch pipeline and integrating JaCoCo for automated coverage validation.

Core Team Members – Other Developers

Implemented assigned functional modules according to their respective Epics and User Stories, including user registration, authentication, and scroll management features. Each member maintained testing accountability for their code.

This role allocation balanced project management responsibilities and technical development, enabling the team to maintain both delivery speed and software quality throughout the sprint.

5.2 Communication and Collaboration Tools

The team continued to hold offline meetings for key decisions and used WeChat for daily communication. GitHub Projects tracked progress of User Stories, and Jenkins monitored build results in real time. This tool set proved efficient and transparent.

5.3 Group Discussion

During the third sprint phase, our team held several offline meetings to discuss various matters, including design decisions, progress reports, and task coordination. We ensured that each user story was implemented according to a unified standard and was consistent with the overall project goals.

At the beginning of the sprint, we reviewed the product backlog and determined the main development goals based on dependencies and priorities, including administrator users and

bookmark functionality. During the meeting, team members proposed different technical solutions, compared various design ideas, and clarified the interfaces between modules.

In addition, we also added a help command to enhance user-friendliness.

After each major milestone was completed, we would hold regular review meetings. Each member would introduce the functions they had completed (such as new commands or optimised interface interaction processes), and receive feedback from other members.

5.4 Individual Contributions

Xuejian Fang implemented the administrator commands for viewing and creating users (US-B1, US-B2) and enhanced the GitHub Release workflow so that each release automatically includes a version tag. He also coordinated CI/CD integration as the Scrum Master.

Zhiwei Cheng developed and tested the delete-user and role-assignment commands (US-B3, US-B4), ensuring reliable permission control and safe deletion mechanisms. He also contributed to integration testing and peer reviews.

Xuanhao Yu implemented the scroll statistics module (US-B5) and the demo artefact packaging (US-F5), verifying that the CLI JAR could run independently without an IDE. He also checked coverage reports and tested system stability.

Boning Zhao completed the bookmark module (US-BM1) and designed the enhanced Help command (US-UX1), allowing users to type help to list all available commands or help <command> to view detailed flag usage. He also maintained documentation and wrote clear User Stories on GitHub.

5.5 Additional Notes and Reflections

Sprint 3 marked the successful completion of the VSAS project, with all planned features delivered on time and thoroughly tested.

Throughout the sprint, the team improved in communication, coordination, and workflow automation. The experience highlighted the importance of starting the project early to leave sufficient time for debugging and integration.

Additionally, maintaining frequent communication and progress synchronisation among members proved essential for avoiding last-minute issues and ensuring smooth delivery.